

Revisiting Chamley and Aiyagari:

Existence and Uniqueness of Steady States in Ramsey Taxation Models

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Roadmap

Introduction

The Bridge Model: Complete & Incomplete Markets

Incomplete Markets Model Solution

Conclusion

Literature: From Chamley–Judd Onwards

Chamley (1986) & Judd (1985) — complete markets:

- Assume the steady state exists.
- $\tau_k = 0$ in the steady state.
- **Infinitely many** steady states, depending on initial debt.

Can these results be generalized?

- $\tau_k > 0$: Aiyagari (1995), incomplete markets.
- **Existence:** Steady state may not exist — Straub–Werning for Judd (no debt) when $\sigma > 1$; still exists for Chamley with debt.
- **Uniqueness:** Unique steady state in incomplete-market models (AHHW, 2026).

Key Gap in the Literature

Scattered and seemingly conflicting results across models — a **unified framework** is needed to connect them and understand them.

This Paper: A Bridge Model

This paper offers a **bridge model** to:

- **Connect** the seemingly conflicting results across models, with a more generalized model and unified logic.
- **Understand** existence and uniqueness across models — not only of theoretical interest, but also crucial for quantitative **correctness, efficiency, and feasibility**.
- **Inform** when one should or should not assume existence and uniqueness — in which models, under which parameters, and with which policy tools.

Judd (1985) — No Government Debt

- Judd (1985): representative agent, no government debt, $B = 0$.
- Ramsey planner maximizes lifetime utility, choosing $\{\tau_k, \tau_n\}$ subject to government budget and competitive equilibrium.

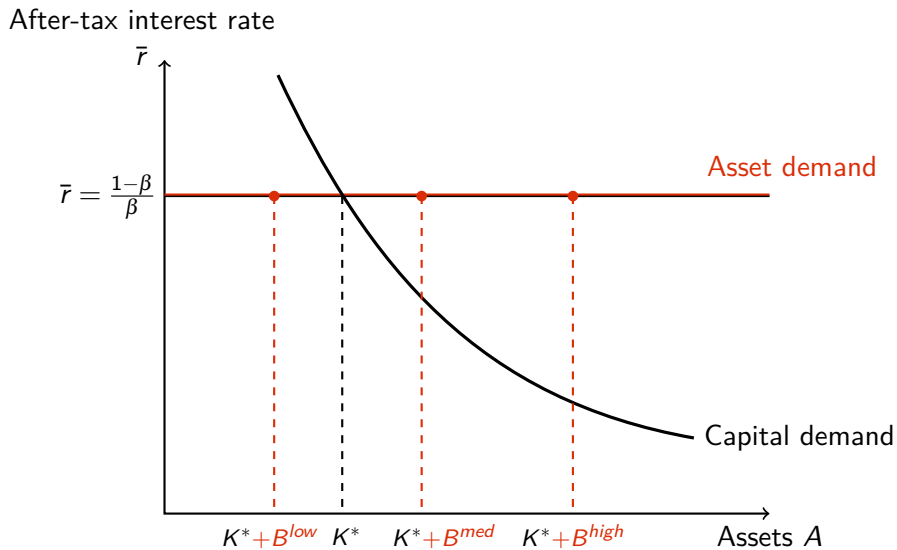
Straub–Werning (2020, AER):

- If $IES < 1$ (i.e., $\sigma > 1$): **steady state does not exist**.
 - With strong savings motives, capital accumulates without bound.
 - Household savings $\rightarrow$ capital stock $\rightarrow$ interest rate $\rightarrow$ more savings.
 - Without debt as a buffer, savings drive unbounded capital accumulation.
- If $IES \geq 1$ (i.e., $\sigma \leq 1$): steady state exists and capital tax is zero (Chamley–Judd result recovered).

Chamley (1986) — With Government Debt

- Chamley (1986): representative agent, **with government debt** B .
- Government debt absorbs household savings — Straub–Werning logic does not apply.
- Steady state **exists** (for any IES, as long as initial debt not too high).
- But **infinitely many steady states**: any level of B is consistent with an optimal SS.
- Intuition: at SS, flat asset demand: any B can be consistent with after-tax interest rate $= 1/\beta - 1$.

Chamley (1986): No Trade-off — Flat Asset Demand



Incomplete Markets

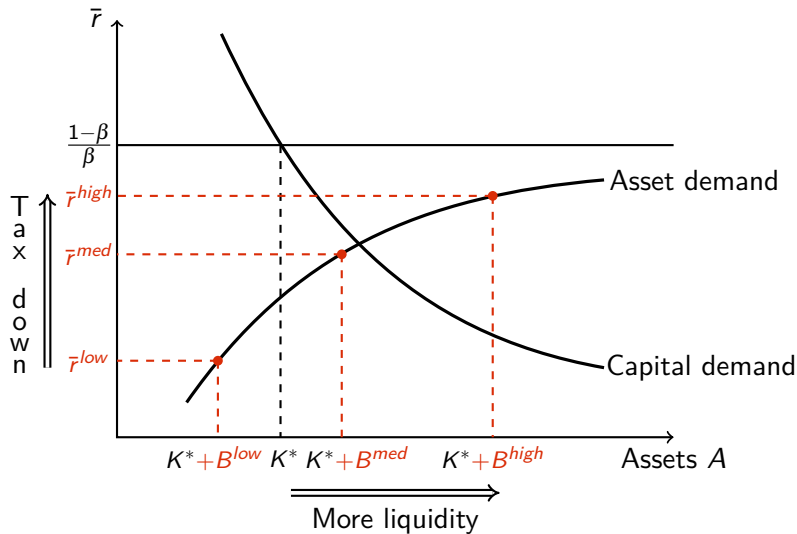
- Aiyagari (1995): **heterogeneous agents**, idiosyncratic income risk, borrowing constraints.
- Precautionary savings $\Rightarrow$ **upward-sloping asset demand**.

Acikgoz, Hagedorn, Holter, Wang (2026):

- Upward-sloping demand pins down unique $(\bar{r}, B)$ pair.
- Trade-off: more debt (liquidity) requires higher $\bar{r} \Rightarrow$ lower τ_k , higher τ_n .
- If Ramsey SS exists $\Rightarrow$ **unique and independent of initial debt**.

Incomplete Markets: The Trade-off

After-tax interest rate



Increasing asset demand curve in incomplete market models helps with uniqueness, but important questions still remain:

Remaining questions:

1. Can **uniqueness** be proved analytically in models with an increasing asset demand curve?
 2. Can the increasing asset demand curve help answer the question on **existence**?
- Heterogeneous agent models are **computationally complex**.
 - Hard to characterize Lagrange multipliers analytically.
 - Existence assumed and verified numerically — no analytical proof.

This Paper: A Tractable Bridge Model

This Paper

Provide a tractable representative agent framework that delivers **clean analytical results** on existence and uniqueness, bridging the RA and HA literatures.

Idea: Add direct utility from asset holdings to a representative agent model.

$$U(c, n, a') = u(c, n) + v(a'), \quad v(a) = \chi_a \frac{a^{1-\sigma_a}}{1-\sigma_a}$$

- $\chi_a \rightarrow 0$: $v(a) \rightarrow 0$, model collapses to Chamley $\Rightarrow$ flat asset demand.
- $\chi_a > 0$: $v'(a) > 0 \Rightarrow$ upward-sloping asset demand, mirroring incomplete markets.
- **Tractability:** analytical results on existence and uniqueness.

This Paper: Main Findings & Contributions

Main Contributions:

1. Existence depends on **curvature** σ_a (not scale χ_a).
2. Uniqueness holds even for $\chi_a \rightarrow 0$: any $\chi_a > 0$ eliminates multiplicity.
3. Unified logic and analytical solution for:
 - Complete markets: Chamley, Straub–Werning.
 - Incomplete markets: Aiyagari, AHHW.

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Model: Household

Representative household, **GHH preferences** with utility from assets:

$$\max_{\{c_t, n_t, a_{t+1}\}} \sum_{t=0}^{\infty} \beta^t \left[\frac{\tilde{c}_t^{1-\sigma}}{1-\sigma} + \chi_a \frac{a_{t+1}^{1-\sigma_a}}{1-\sigma_a} \right],$$
$$\tilde{c}_t = c_t - \chi \frac{n_t^{1+1/\phi}}{1+1/\phi}$$

subject to budget constraint:

$$c_t + a_{t+1} = (1 + \bar{r}_t) a_t + \bar{w}_t n_t + T_t$$

- $\bar{r}_t = (1 - \tau_{k,t}) r_t$: after-tax return on assets.
- $\bar{w}_t = (1 - \tau_{n,t}) w_t$: after-tax wage.
- **No capital** in baseline ($Y = Z \cdot N$); adding capital does not change SS results (modified golden rule pins down K/N).

Model: Government and Technology

Government budget constraint:

$$G + T_t + \bar{r}_t B_t = \tau_{n,t} w_t N_t + B_{t+1} - B_t$$

Technology (no capital baseline):

$$Y_t = Z \cdot N_t, \quad w_t = Z, \quad r_t = \frac{1}{\beta} - 1$$

- Planner chooses after-tax prices $\{\bar{w}_t, \bar{r}_t\}$.
- Transfers T_t exogenous (not optimally chosen) — key for SS existence.
- Market clearing: $B_t = a_t$ (bonds = assets).
- GHH utility $\Rightarrow$ no wealth effect on labor: $n_t = (\bar{w}_t/\chi)^\phi$.

Ramsey Lagrangian

Choice variables: $\{a_{t+1}, \bar{w}_t, \bar{r}_t\}_{t=0}^{\infty}$. Constraints: EE + GovBC (no HHBC).

Government Budget Constraint (GovBC):

$$G_t + T_t = \underbrace{(Z_t - \bar{w}_t) n_t}_{\text{labor tax}} + \underbrace{a_{t+1} - (1 + \bar{r}_t) a_t}_{\text{net borrowing}}$$

Lagrangian (with θ_{t+1} on EE, γ_t on GovBC):

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \beta^t & \left\{ u(c_t, n_t) + v(a_{t+1}) \right. \\ & + \theta_{t+1} \underbrace{\left[u_c(c_t, n_t) - \beta(1 + \bar{r}_{t+1})u_c(c_{t+1}, n_{t+1}) - v_a(a_{t+1}) \right]}_{\text{Euler equation}} \\ & \left. + \gamma_t \underbrace{\left[Z_t n_t + a_{t+1} - \bar{r}_t a_t - \bar{w}_t n_t - a_t - G_t - T_t \right]}_{\text{GovBC}} \right\} \end{aligned}$$

Rearranged Lagrangian

Define $\lambda_t \equiv -\theta_t$:

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, n_t) + v(a_{t+1}) + u_c(c_t, n_t) [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] \right. \\ \left. + v_a(a_{t+1}) \lambda_{t+1} + \gamma_t [Z_t n_t + a_{t+1} - \bar{r}_t a_t - \bar{w}_t n_t - a_t - G_t - T_t] \right\}$$

Euler Equation (constraint $\partial \mathcal{L} / \partial \lambda_{t+1} = 0$):

$$u_c(c_t, n_t) = \beta(1 + \bar{r}_{t+1}) u_c(c_{t+1}, n_{t+1}) + v_a(a_{t+1})$$

Modified Euler Equation (ModEE)

Rearranged Lagrangian:

$$\mathcal{L} \supset \beta^t \left\{ u_t + v(a_{t+1}) + u_{c,t} [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] + v_a(a_{t+1})\lambda_{t+1} + \gamma_t[\text{GovBC}] \right\}$$

ModEE ($\partial \mathcal{L} / \partial a_{t+1}$):

$$\begin{aligned} u_{cc,t} [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] &= \beta(1 + \bar{r}_{t+1}) u_{cc,t+1} [\lambda_{t+1}(1 + \bar{r}_{t+1}) - \lambda_{t+2}] \\ &\quad + v_{aa,t+1} \lambda_{t+1} + \gamma_t - \beta \gamma_{t+1} (1 + \bar{r}_{t+1}) \end{aligned}$$

FOC for $\bar{r}_t$

Rearranged Lagrangian:

$$\mathcal{L} \supset \beta^t \left\{ u_t + v(a_{t+1}) + u_{c,t} [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] + v_a(a_{t+1})\lambda_{t+1} + \gamma_t[\text{GovBC}] \right\}$$

FOC $\bar{r}_t$ ($\partial \mathcal{L} / \partial \bar{r}_t$):

$$\gamma_t = u_c(c_t, n_t) \frac{\lambda_t}{a_t} + u_c(c_t, n_t) + u_{cc}(c_t, n_t) [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}]$$

FOC for $\bar{w}_t$

Rearranged Lagrangian:

$$\mathcal{L} \supset \beta^t \left\{ u_t + v(a_{t+1}) + u_{c,t} [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] + v_a(a_{t+1})\lambda_{t+1} + \gamma_t[\text{GovBC}] \right\}$$

FOC $\bar{w}_t$ ($\partial \mathcal{L} / \partial \bar{w}_t$):

$$\gamma_t = \gamma_t \left(\frac{Z_t}{\bar{w}_t} - 1 \right) \phi + u_c(c_t, n_t) + u_{cc}(c_t, n_t) [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}]$$

Subtracting FOC $\bar{r}_t$ – FOC $\bar{w}_t$ (for $t > 0$):

$$\gamma_t = \frac{u_c(c_t, n_t) \lambda_t}{\phi(Z_t/\bar{w}_t - 1) a_t}$$

Steady State: The Key Condition

At the steady state (drop time subscripts), combining FOCs:

Step 1: FOC $\bar{r}$ + ModEE:

$$(\gamma - u_{cc}\lambda\bar{r})\frac{v_a}{u_c} = -v_{aa}\lambda$$

Step 2: Substitute FOC $\bar{r}$, cancel u_c :

$$u_c\left(\frac{\lambda}{a} + 1\right) = \sigma_a \frac{u_c\lambda}{a}$$

Step 3: u_c cancels completely:

$$\frac{\lambda}{a} + 1 = \sigma_a \frac{\lambda}{a}$$

Key Steady-State Condition

$(\sigma_a - 1)\frac{\lambda}{a} = 1$ — determines existence and pins down λ/a uniquely.

Existence: Depends on Curvature σ_a

Proposition 1 (Existence)

- If $\sigma_a > 1$: **steady state exists** with $\lambda > 0$ (planner wants more savings).
Speed of convergence $\propto (\sigma_a - 1) v_a / u_c$.
- If $\sigma_a = 1$: **no steady state** (λ does not converge).
- If $\sigma_a < 1$: **no steady state** ($\lambda < 0$, does not converge).

Intuition:

- $\sigma_a > 1$: $v(a)$ is sufficiently concave $\Rightarrow$ self-limiting savings.
- $\sigma_a = 1$ (log): savings motives never self-limit (mirrors Straub–Werning); λ diverges.
- $\sigma_a < 1$ (v closer to linear): high incentive to save, savings explode.

Uniqueness: Independence from χ_a

Proposition 2 (Uniqueness)

The optimal transition converges to a **unique steady state**, independent of initial conditions such as initial debt B_0 .

Uniqueness holds even as $\chi_a \rightarrow 0$: real variables $\bar{w}, c, n, \gamma$ are **independent of χ_a** .

Proof sketch: ModEE transforms to:

$$\phi \left(\frac{Z}{\bar{w}} - 1 \right) + \sigma \frac{c - \bar{w}n - T}{\tilde{c}} = \sigma_a$$

- No χ_a : $\bar{w}$ determined solely by $\sigma_a, \sigma, \phi, Z, G, T$.
- From $\bar{w}$ recover n, c, γ — all independent of χ_a .
- Nominal variables $a, \bar{r}, \lambda$ do depend on χ_a .

Economic interpretation

Chamley multiplicity is non-generic: any $\chi_a > 0$, no matter how small, eliminates the continuum of steady states and selects a unique one.

Analytical Results: Special Cases

Proposition 3 (Special Case with Analytical Solution)

If $\phi = 1$ and $G = T = 0$, the simplified ModEE reduces to a **quadratic equation** in $\bar{w}$.

If further $\sigma = \sigma_a$, it reduces to a **linear equation**:

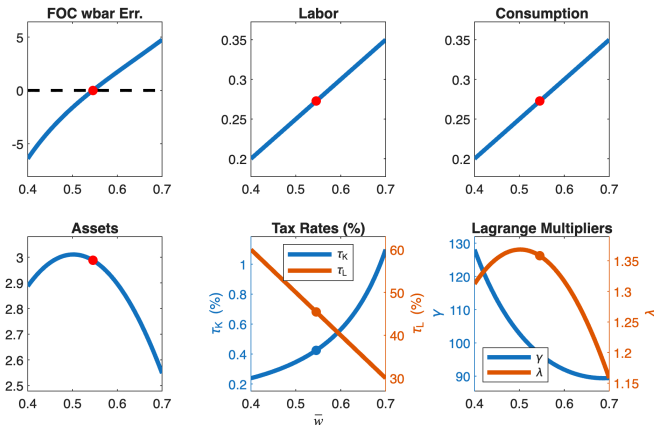
$$\tau_l = \frac{1}{\sigma - 1}$$

Steady State Algorithm

Key innovation: Solve for $\bar{w}$ directly, without solving for γ, λ, a .

1. **Guess** $\bar{w}$.
2. Labor: $n = (\bar{w}/\chi)^\phi$; consumption: $c = ZN - G$.
3. $\bar{r}a$ from household budget: $\bar{r}a = c - \bar{w}n - T$.
4. a and $\bar{r}$ from EE: $v_a(a) = \beta(r - \bar{r}/a \cdot a) u_c$.
5. **Verify** $\bar{w}$ via simplified ModEE: $\phi\left(\frac{Z}{\bar{w}} - 1\right) + \sigma \frac{c - \bar{w}n - T}{\bar{c}} = \sigma_a$
6. Update $\bar{w}$ until convergence; recover γ, λ after.
 - Single equation in single unknown $\bar{w}$; χ_a does not enter the loop.

Numerical Illustration: Asset Demand Curves



- Upward-sloping asset demand (for $\sigma_a > 1$): intersection is unique.
- As $\chi_a \rightarrow 0$: curve flattens but unique SS preserved (Prop. 2).
- χ_a shifts the nominal level of a , but not $\bar{w}$ or τ_l .

Connecting to the Literature: Summary

Model	Debt	Asset demand	SS result
Judd (1985), $\sigma > 1$	No	Flat	Non-existent
Judd (1985), $\sigma \leq 1$	No	Flat	Exists, ∞ many
Chamley (1986)	Yes	Flat	Exists, ∞ many
Aiyagari (1995), AHHW (2026)	Yes	Upward-sloping	Exists, unique
This paper: $\chi_a > 0$, $\sigma_a > 1$	Yes	Upward-sloping	Exists, unique
This paper: $\chi_a \rightarrow 0$	Yes	$\rightarrow$ Flat	Unique $\rightarrow \infty$, Chamley
This paper: $\sigma_a \leq 1$	Yes	Upward-sloping	Non-existent

Warning: Optimal Transfers Eliminate the Steady State

What if the Ramsey planner can freely choose lump-sum transfers T_t ?

Proposition (First Best with Transfers)

*If optimal lump-sum transfers are available, the Ramsey planner can achieve the **first-best allocation**:*

$$\bar{w} = Z(\tau_n = 0), \quad \bar{r} = \frac{1}{\beta} - 1 (\tau_k = 0), \quad a = v_a^{-1}(0) \doteq \bar{a}$$

- $v(a)$ has satiation point $\bar{a}$ (e.g. quadratic): first best at $a = \bar{a}$ — interior SS exists.
- $v(a)$ strictly increasing (e.g. CRRA): **no interior SS** — planner keeps raising a indefinitely.

Implication

With flexible transfers, SS may not exist if more assets always increase utility.

Warning: Be Careful About Transfers in Incomplete Market Models

- Bhandari, Evans, Golosov, Sargent (2017): with flexible transfers, planner front-loads to remove the borrowing constraint $\Rightarrow$ **Ricardian equivalence restored**, debt indeterminate.
- AHHW: **no borrowing constraint** $\Rightarrow$ Ramsey SS does not exist (λ_t diverges).
- Models with transfers as a Ramsey instrument may not converge, or have an indeterminate SS.
- Restrict instrument set to distortionary taxes (and possibly fixed transfers) to keep borrowing constraint occasionally binding.

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Incomplete Market Model: AHHW

Key differences from the bridge model:

- **Heterogeneous agents:** continuum of households, idiosyncratic productivity shocks e_{it} .
- Borrowing constraint $a_{t+1} \geq \underline{a}$.
- **Capital** explicitly in the model: $Y = K^\alpha (eN)^{1-\alpha}$.

Same Lagrangian approach: FOCs for prices $\{\bar{r}_t, \bar{w}_t\}$, Lagrange multipliers $\{\gamma_t, \lambda_t\}$.

Ramsey Lagrangian: Bridge Model vs. AHHW

Bridge model:

$$\mathcal{L} \supset \beta^t \left\{ u(c_t, n_t) + v(a_{t+1}) + u_{c,t} [\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}] + v_a(a_{t+1}) \lambda_{t+1} + \gamma_t [\text{GovBC}] \right\}$$

AHHW (incomplete markets):

$$\mathcal{L} = \int \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ u(c_t, n_t) + u_c(c_t, n_t) (\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}) \right. \\ \left. + \gamma_t (F(K_t, N_t) - \delta K_t + B_{t+1} - G_t - T_t - (1 + \bar{r}_t) B_t - \bar{r}_t K_t - \bar{w}_t N_t) \right\} d\mu_0$$

- **Blue**: same structure. In AHHW, $\lambda_t = \theta_t - \eta_t(a_t + \underline{a})$ absorbs the borrowing-constraint wedge.
- **Red** $v_a(a')\lambda'$ (bridge) $\leftrightarrow \mathbb{E}_0$ (AHHW): both are the source of curvature, generating upward-sloping asset demand.
- **Green**: GovBC; richer in AHHW (includes capital).

AHHW: First-Order Conditions

EE $(\partial \mathcal{L} / \partial \lambda_{t+1})$:

$$u_{c,t} \geq \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t[u_{c,t+1}], \quad \text{with equality if } a_{t+1} > -\underline{a}$$

ModEE $(\partial \mathcal{L} / \partial a_{t+1}, \text{ if } a_{t+1} > -\underline{a}; \text{ otherwise } \lambda_{t+1} = 0)$:

$$u_{c,t} + u_{cc,t}(\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}) = \beta(1 + \bar{r}_{t+1}) \mathbb{E}_t[u_{c,t+1} + u_{cc,t+1}(\lambda_{t+1}(1 + \bar{r}_{t+1}) - \lambda_{t+2})] \\ + \beta \gamma_{t+1} (F_K(K_{t+1}, N_{t+1}) - \delta - \bar{r}_{t+1})$$

FOC B_{t+1} $(\partial \mathcal{L} / \partial B_{t+1})$:

$$\gamma_t = \beta (F_K(K_{t+1}, N_{t+1}) + 1 - \delta) \gamma_{t+1}$$

FOC $\bar{r}_t$ $(\partial \mathcal{L} / \partial \bar{r}_t)$:

$$\gamma_t A_t = \mathbb{E}_t[u_{c,t} \lambda_t + a_t (u_{c,t} + u_{cc,t}(\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}))]$$

FOC $\bar{w}_t$ $(\partial \mathcal{L} / \partial \bar{w}_t)$:

$$\gamma_t N_t = \gamma_t (F_{N,t} - \bar{w}_t) \frac{\partial N_t}{\partial \bar{w}_t} + \mathbb{E}_t \left[e_t n_t u_{c,t} + \left(\frac{\partial c_t}{\partial \bar{w}_t} u_{cc,t} + \frac{\partial n_t}{\partial \bar{w}_t} u_{cn,t} \right) (\lambda_t(1 + \bar{r}_t) - \lambda_{t+1}) \right]$$

AHHW: Theoretical Results

Theorem (Modified Golden Rule)

The long-run capital stock satisfies $F_K(\bar{K}, \bar{N}) = 1/\beta$.

Theorem (Independence of Initial Conditions)

*The long-run values of $\bar{B}/Y$, $\bar{\tau}_n$, and $\bar{\tau}_k$ are **generically independent** of initial debt and capital.*

Intuition:

- Upward-sloping asset demand $\Rightarrow$ finite number of SS solutions.
- Berge's Maximum Theorem $\Rightarrow$ optimal policy is continuous in initial conditions.
- A discrete continuous function is constant.

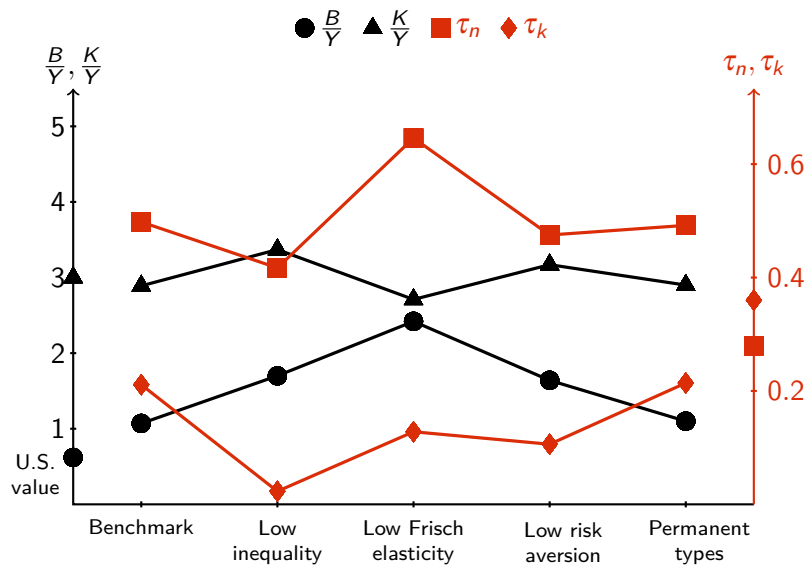
AHHW: Calibration

Param.	Value	Description	Source/Target
<i>Exogenous Parameters</i>			
σ	2	Coeff. risk aversion	
ϕ	1	Frisch elasticity	
α	0.36	Capital share	
δ	0.08	Depreciation rate	
τ_l	28%	Labor inc. tax	Trabandt-Uhlig (2011)
τ_k	36%	Capital inc. tax	Trabandt-Uhlig (2011)
B/Y	62%	Debt-to-GDP	Holter et al. (2015)
G/Y	7.3%	Gov. spending	Prescott (2004)
<i>Calibrated Parameters</i>			
ρ	0.94	Persistence of prod. shock	$a_{90}/a_{50} = 7.55$
σ_u	0.18	S.D. of prod. shock	$\text{var}(\log y) = 1.3$
β	0.96	Discount rate	$K/Y = 3$
χ	2.56	Disutility from labor	$\bar{n} = 0.33$

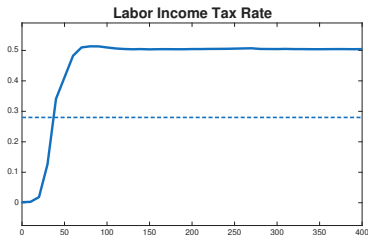
AHHW: Optimal Steady State

	Optimal SS (1)	U.S. Calibration (2)
τ_l	49.8%	28.0%
τ_k	21.5%	36.0%
T/Y	21.1%	14.1%
B/Y	1.07	0.62
K/Y	2.89	3.00
$\int n$	0.23	0.33
$N = \int en$	0.31	0.46
$\text{var}(\log(y + T))$	0.30	0.51
$\text{var}(\log(y + T + \bar{r}a))$	0.35	0.52
Coeff. var. a	1.37	1.52
Coeff. var. c	0.64	0.80

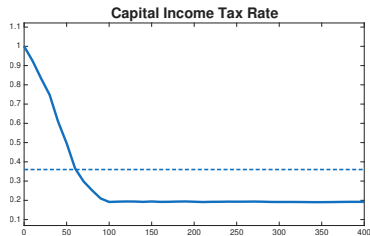
AHHW: Robustness of Steady State



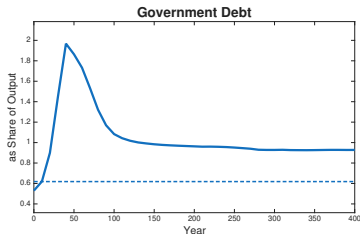
AHHW: Optimal Transition Path



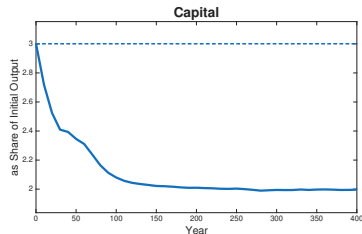
Labor tax τ_n



Capital tax τ_k



Debt-to-output B/Y



Capital stock K

AHHW: Robust Conclusions

- High government debt in the long run.
 $\hookrightarrow$ Tax distortions do not tightly bound the level of debt.
- Low capital income tax.
 $\hookrightarrow$ Large debt requires after-tax rate close to $1/\beta - 1$.
- High labor tax.
 $\hookrightarrow$ High interest payments must be financed.
- Along the transition:
 Capital tax is **high initially** (front-load capital taxation),
 then declines to its long-run level.
 Labor tax rises over time.
- Welfare gain: $\varphi = 2.59\%$ consumption equivalents relative to U.S. calibration.

This paper so far: GHH utility, no capital.

- **Add capital:** K/N satisfies modified golden rule; results carry over with B absorbing savings beyond K^* .
- **Different utility:** Extend to separable $U(c, n) = u(c) - v(n)$.
 - Auclert et al. (2024): SS existence depends on labor supply response to wealth.
 - GHH eliminates wealth effect $\Rightarrow$ SS exists; separable utility $\Rightarrow$ SS may fail.
- **Analytical proof via bridge model:** Use tractability to prove Auclert et al. analytically — identify exact role of wealth effect in preventing convergence.

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Assets-in-Utility Bridge Model

- Existence of Ramsey SS requires $\sigma_a > 1$ (curvature of $v(a)$).
- Uniqueness holds for any $\chi_a > 0$, even $\chi_a \rightarrow 0$.
- Chamley multiplicity is **non-generic**: infinitesimally small χ_a restores uniqueness.
- Clean analytical results; closed-form benchmark for numerical validation.

AHHW Optimal Taxation

- Modified golden rule: $F_K = 1/\beta$ in the long run; unique SS independent of initial conditions.
- **Capital tax** τ_k : front-loaded, then low in long run (21% vs. 36% in U.S.).
- **Labor tax** τ_n : rises to 50% in long run (vs. 28% in U.S.).
- **Debt** B/Y : rises to 107% in long run (vs. 62% in U.S.).

Unified message

The upward slope of the asset demand curve is the key for uniqueness.

Appendix

Algorithm Details: Transition Path (AHHW)

1. Given after-tax prices $\{\bar{w}_t, \bar{r}_t\}$, solve for household policy functions.
2. Simulate economy forward from initial distribution μ_0 ; compute $\{K_t, B_t, N_t\}$.
3. Compute capital and debt **backward** to satisfy resource constraints.
4. Solve for Lagrange multipliers $\{\gamma_t, \lambda_t\}$ **backward**.
5. Simulate economy forward; check FOCs w.r.t. prices.
6. Update prices until convergence.
 - Each period = 10 years (slow distribution convergence).
 - Exploit independence result: SS computed first, independently of initial conditions.